

Ahmed Ikram | Software Engineering

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PROFILE

4th-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Builds production-ready software end-to-end across React/React Native frontends, FastAPI/Flask backends, PostgreSQL data models, and cloud infrastructure on AWS and Azure - with multi-service Docker deployments, OIDC-authenticated CI/CD, and 1,400+ automated tests gating production changes. Seeking a graduate Software Engineering or Full-Stack role to contribute strong backend design, cloud deployment, and engineering discipline in a team shipping real products.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

DevSync Project Management Platform with GitHub Integration | React, Tailwind, Python, Flask, PostgreSQL, Socket.io, GitHub OAuth, AWS, Docker, GitHub Actions, CI/CD

- React, Tailwind frontend, Flask & Gunicorn REST API over a 12-table indexed PostgreSQL schema, and a JWT-authenticated Socket.IO layer with project-scoped rooms - live task updates, notifications, and dashboard refreshes broadcast only to members viewing the affected project, with no page refresh required; GitHub activity fetched via GraphQL API to reduce latency when creating charts and reports
- Implemented three-tier RBAC (Developer / Team Leader / Admin) via role decorators on every route; JWT in HTTP-only cookies; GitHub OAuth 2.0 handled server-side with tokens never sent to the browser; bidirectional task ↔ issues/PR sync means closing a GitHub issue automatically updates the linked task in DevSync
- Production-like AWS infrastructure: ECS Fargate behind an ALB, RDS in a private subnet, and S3, CloudFront & ACM for the React SPA - frontend deployment blocked until ECS health checks pass to prevent API/UI version mismatch; 1,452 automated tests at 85%/85% coverage hard-gate every PR via 7 path-aware CI jobs (lint: ruff + ESLint, security: pip-audit + npm audit, Postgres 15 integration, Cypress E2E), OIDC federation for CI/CD with no static credentials, Docker layer caching for 70%+ faster builds, secrets injected at runtime via AWS Secrets Manager and per-task IAM roles scoped to least-privilege Dependabot + CodeQL on weekly cadence

ATM Log Aggregation, Analysis & Diagnostics Platform | Python, FastAPI, React, PostgreSQL, Kafka, Redis, nginx, Docker, Terraform, AWS, SageMaker

- Backend engineering lead on a 7-person Agile team delivering a diagnostics platform for NCR Atleos; independently architected a 10-service Dockerised platform, full AWS production deployment, and 3-pipeline CI/CD system - all built post-deadline on top of the team-delivered foundation
- Architected a 10-service system with health-check cascading startup: FastAPI REST API (30 endpoints, 6 routers, JWT + Redis blacklist auth), Kafka Kraft consumer with hybrid Redis+LRU deduplication and a Redis Stream DLQ, 8 Redis distributed patterns and a React frontend via multi-stage Docker build (~25MB nginx-alpine final image); deployed on AWS via Terraform (10 modules, 118 resources) - multi-AZ VPC (2 public + 2 private subnets), ECS Fargate (2 services × 2 tasks across AZs) behind an ALB, EC2 t3.medium hosting Kafka/Redis/ChromaDB/Ollama in private subnets, CloudFront serving the React SPA from S3, 8 secrets injected at container start via Secrets Manager, and 6 least-privilege IAM roles
- Built a 3-pipeline CI/CD system: CI runs ruff/mypy/bandit lint, checkov IaC compliance, and 1,402 tests (923 pytest at 82% coverage, 394 vitest, 10 Playwright E2E, 75 Terraform assertions) in parallel on every PR; CD-SHOULD-DEPLOY uses dorny/paths-filter to skip infrastructure jobs on documentation-only changes; CD applies Terraform plan/apply across all 10 modules and triggers a zero-downtime ECS rolling update on merge to main - all GitHub Actions pipelines authenticate to AWS via OIDC AssumeRole with no long-lived credentials stored anywhere

StockLens FinTech Spending & Investment App | *React Native, Python, FastAPI, PostgreSQL, Redis, Terraform, Docker, AWS*

- Biometric authentication and real-time portfolio views, async FastAPI backend exposing 59 REST endpoints across 14 routers with JWT auth, bcrypt password hashing, slowapi rate limiting and tenacity retry on the agent endpoint
- Native Rust extension (PyO3/Maturin), with 12 exported exported functions compiled to a wheel and cross-built for ARM64 via QEMU - called directly by both the training pipeline and live API with zero Python-side overhead
- Deployed on AWS via Terraform (14 modules, 193 resources): multi-AZ VPC with three-tier network isolation (Internet → WAF-protected ALB → ECS Fargate ARM64 → private RDS/Redis), Multi-AZ RDS and TLS/AUTH-enabled ElastiCache Redis, CPU/RPS-triggered auto scaling (min=2, max=6), and zero-downtime rolling deploys gated on ECS health checks passing before traffic cutover
- 9-parallel-job CI pipeline (lint, TypeScript, Jest, Rust clippy/cargo test, pytest at a 90% coverage gate, Docker validation, IaC scanning, secrets scanning) and a 7-stage OIDC-authenticated CD pipeline (build → push → scan → plan → manual review → apply) with zero long-lived AWS credentials, gated by Checkov, tfsec, Gitleaks, and Trivy

W3C Web Logs ETL Pipeline | *Apache Airflow, Python, Databricks, dbt, Apache Spark, PySpark, Azure SQL, Terraform, Docker, GitHub Actions, Power BI, Grafana, Prometheus*

- Built a serverless Databricks medallion pipeline (Bronze → Silver → Azure SQL → dbt → Power BI) ingesting 93 IIS log files (155,380 rows, 3 dropped) across 7 @dlt.expect_or_drop quality gates; a custom UDF handles 14- vs 18-field format detection dynamically; 4 Airflow DAGs orchestrate the full pipeline with zero data-processing logic in DAG files - layers decoupled via Airflow Datasets, no ExternalTaskSensor
- Provisioned 30+ Azure resources across 2 Terraform configurations including the full OIDC auth chain (Azure AD app, service principal, federated credential, role assignment) via a single terraform apply; 7-workflow CI/CD pipeline with 4 parallel jobs and a rollback workflow that re-applies Terraform from the previous commit on demand — zero static credentials anywhere
- Delivered a 7-page Power BI dashboard (156K requests, 88 countries, P95 1,149ms, 38% bot traffic, 9.7% 404 rate) from 18 dbt-generated CSV exports via a semantic contract; observability: custom Python freshness probe → 3 Grafana dashboards (23 panels) → 10 alert rules across 3 severity-tiered action groups with Slack notifications; 613 tests across 5 tiers including dual-dialect dbt CI against side-by-side PostgreSQL and SQL Server containers

TECHNICAL SKILLS

Languages	Python · TypeScript · JavaScript · SQL · Bash · Rust
Frontend & Mobile	React · React Native · Expo · Tailwind · Vite · Chart.js · Socket.IO
Backend & APIs	FastAPI · Flask · Node.js · GraphQL · Kafka · Gunicorn · Airflow · dbt · Pydantic
Security	JWT · OAuth 2.0 · RBAC · AWS Secrets Manager · bcrypt
Databases	PostgreSQL · Redis · SQLite · MySQL · MongoDB · Alembic
Cloud & DevOps	AWS (ECS Fargate, EC2, S3, RDS, Elasticache, WAF, IAM, CloudFront, Sagemaker, Bedrock, Secrets Manager) · Azure (Databricks, Azure SQL, ADLS) · Terraform · Docker · nginx · GitHub Actions · Linux
Infra & Reliability	OIDC Federation · GitHub Actions CI/CD · Checkov · tfsec · Trivy · GitLeaks · QEMU
Observability	Grafana · Prometheus · StatsD · Slack (alert routing)
Testing	Pytest · Vitest · Jest · Playwright · React Testing Library · Cypress · dbt Tests

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>

EXPERIENCE

Beaully Mini Market, Dundee, Scotland <i>Retail Assistant</i>	<i>September 2023 - Present</i>
• Troubleshoot and resolved Point-of-Sale and stock system errors quickly to minimise customer wait time	
Zulus Peri Peri Grill, Dundee, Scotland <i>Front of House Worker</i>	<i>June 2021 - September 2023</i>
• Trained 5 new staff members, helping them get up to speed quickly during busy service periods	